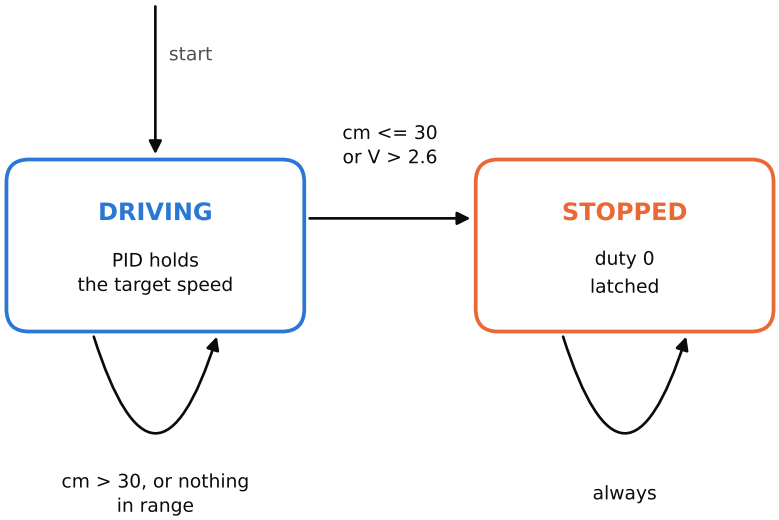


Fig. 9.1. Stop logic of avoid.py, should_stop(), rev d22b77d. Checked once per control loop, about every 82 ms. DRIVING holds the target speed with the Lab 7 PID. Any reading at or under 30 cm, or above 2.6 V (too close to measure), moves it to STOPPED, which is latched until the run ends. The e-stop button is separate and cuts the motor driver in hardware from either state.

| t s | volts | cm | stopped | left rpm | left duty | right rpm | right duty |
|---|---|---|---|---|---|---|---|
| 0.496 | 1.356 | 37.8 | 0 | 69.1 | 0.0885 | 68.0 | 0.0943 |
| 0.583 | 1.408 | 36.3 | 0 | 67.9 | 0.0952 | 67.6 | 0.0994 |
| 0.671 | 1.458 | 35.0 | 0 | 66.3 | 0.103 | 68.1 | 0.1027 |
| 0.757 | 1.516 | 33.6 | 0 | 70.2 | 0.0987 | 68.3 | 0.1061 |
| 0.847 | 1.581 | 32.2 | 0 | 70.2 | 0.1021 | 71.4 | 0.103 |
| 0.935 | 1.648 | 30.8 | 0 | 69.2 | 0.108 | 71.7 | 0.1054 |
| **1.044** | **1.718** | **29.5** | **1** | **70.3** | **0.0** | **72.6** | **0.0** |
| 1.128 | 1.819 | 27.8 | 1 | 60.3 | 0.0 | 55.0 | 0.0 |
| 1.21 | 1.858 | 27.2 | 1 | 40.0 | 0.0 | 38.2 | 0.0 |
| 1.286 | 1.881 | 26.9 | 1 | 20.7 | 0.0 | 18.9 | 0.0 |
| 1.356 | 1.919 | 26.3 | 1 | 4.8 | 0.0 | 3.6 | 0.0 |
| 1.426 | 1.923 | 26.3 | 1 | 0.0 | 0.0 | 0.0 | 0.0 |
| 1.497 | 1.906 | 26.5 | 1 | 0.0 | 0.0 | 0.0 | 0.0 |
| 1.568 | 1.91 | 26.4 | 1 | 0.0 | 0.0 | 0.0 | 0.0 |
| 1.639 | 1.916 | 26.4 | 1 | 0.0 | 0.0 | 0.0 | 0.0 |

Exhibit 9A. One stop, loop by loop. Floor test, step 4, last run, 15 Oct 2026, avoid.py 80 rpm, stop at 30 cm, data in avoid_80_30.csv. Duty drops to 0 in the first loop that reads 30 cm or less (bold). The wheels coast to 0 in 0.38 s while the sensor sees the wall close from 29.5 to 26.3 cm. rpm is measured over the loop before each row.